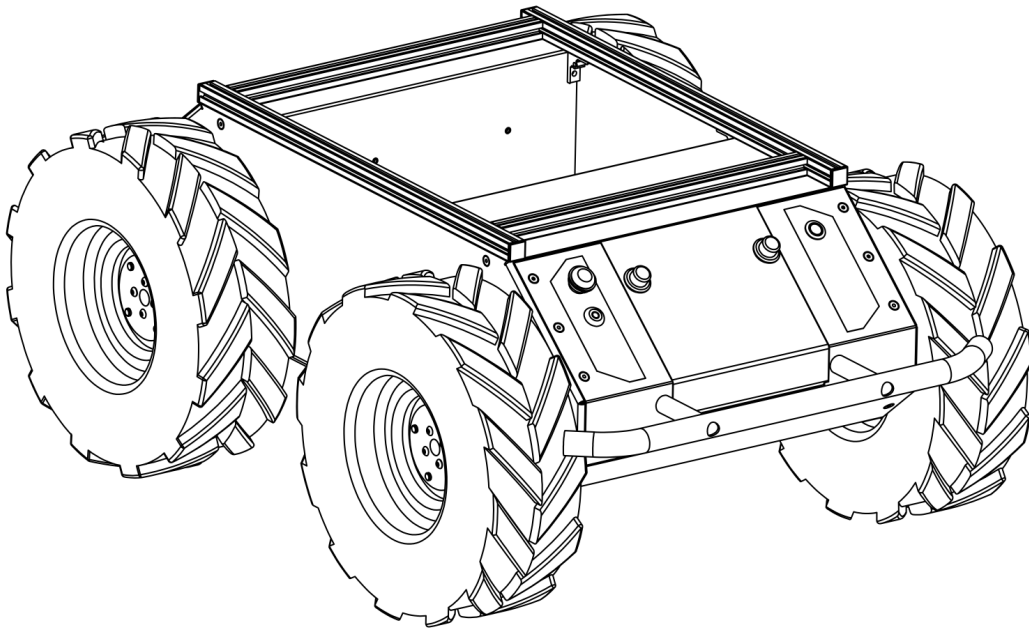


UGA10 HUSKY



UNIQUELY INTEGRATED PLATFORM

CUSTOM ROBOT QUICKSTART GUIDE

1 FROM THE DESK OF THE ROBOTSMITHS

Thank you University of Georgia for your Husky order! As part of the integration, we have prepared this quick reference memo for you and your team detailing your specific package.

2 SAFETY WARNING

Use of an autonomous robot is inherently dangerous. Please take time to locate the Red E-STOP button on the rear of the Husky. The E-STOP and key switch on the rear of the Husky are used to stop the Husky movement, but DO NOT control power to the payloads.

3 SYSTEM OPERATIONAL TIPS

For best battery performance and life, it is advised that the batteries be charged immediately after use. If the vehicle begins halting unexpectedly, doing one or more of the following may improve performance:

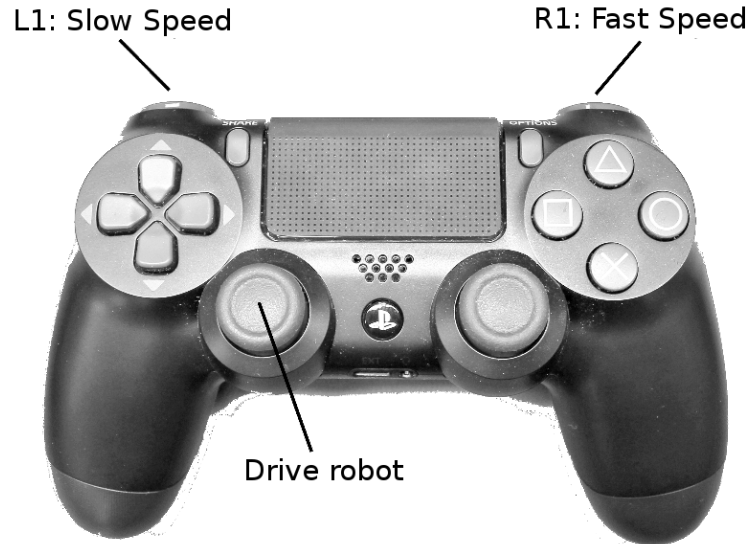
- Reducing motor power draw by lowering acceleration, turning in place less, or limiting the grade of terrain being traversed.
- Reducing peripheral power draw by unplugging or otherwise shutting off devices which are not in use.

Your Husky's battery can be charged by plugging its terminals into the included charger. Simply match red to red and black to black. A light on the top of the charger will illuminate to indicate charging is taking place. When the charging light turns green, the battery is fully charged!

4 GETTING STARTED

Your system has been configured to allow you to get started immediately after receipt. Follow these instructions to get moving:

1. Remove the battery door and connect the battery power cable into the MCU.
2. Turn on the Husky.
3. Release all emergency stop controls on the Husky. The "COMM" light should illuminate green.
4. Press the "PS" button on the gamepad to turn it on. Once Bluetooth is connected you will see a solid light on the top of the controller.
5. Use the gamepad to control Husky's motion.



5 NETWORK INFORMATION

Parameter	Value
Robot Static IP	192.168.131.1
Robot Hostname	cpr-a200-1051
User Name	administrator
User Password	clearpath
ROS Version	Noetic
Serial Number	A200-1051
Velodyne IP	192.168.131.20
SwiftNav Rover IP	192.168.131.31
SwiftNav Base Station IP	192.168.131.30
BulletPlusAC-CAT12 IP	192.168.131.51
BulletPlusAC-CAT12 Username	admin
BulletPlusAC-CAT12 Password	clearpath
Base Station Radio IP	192.168.131.50
Base Station Radio Username	admin
Base Station Radio Password	clearpath
Base Station SSID	UGA10BaseStation

6 BASE STATION

Your robot is equipped with a portable high-powered Wifi Base Station. This base station may be used in your lab or anywhere in the field as a long-distance wifi access point for communicating with your robot. Your robot has been configured to connect with this base station by default.

To set up the Base Station, find a suitable and stable location to set up the base station tripod. Mount the Base Station to the tripod by threading the tripod's mounting screw to the Base Station's threaded mounting hole.

If you are working in a lab environment, you may optionally connect the Base Station's WAN port to a live Internet connection. This will grant the robot (and any other computers connected to the Base Station via wifi) access to the internet.

Some corporate network infrastructures may not permit an unauthorized access point. If this is the case at your location, please contact your IT administrator for assistance with connecting the Base Station to your intranet.

Turn on the Base Station by pressing the silver button located on the side. After a few minutes the Base Station will begin broadcasting its SSID. At this point you may turn on your robot, and connect to the base station using any other external computers. Your computer will be granted an IP address in the same subnet as the Base Station and robot, allowing you to communicate with the robot easily.

The Base Station battery will last for about 6 hours on a full charge. It is a good idea to change the battery as soon as you are finished using the base station, so that you always have a fresh battery ready to go.

Base Station settings may be altered by connecting to it via wifi, then entering its IP address into a web browser. These settings are available in the Network Information section earlier in this document.

7 WIRELESS

The built-in wireless card in your robot is set up as a client, and may be configured to connect to your site wifi infrastructure or other external wireless access point.

To set up the wireless communications on your Husky, you must first establish a wired connection. Using an Ethernet cable, connect your computer to an Ethernet port on the Husky's computer via the top plate, and set a static IP on your computer to 192.168.131.19 (for example). If there are no free ports you may temporarily disconnect one of the payloads. SSH into the robot computer with:

```
ssh administrator@192.168.131.1
```

Enter the login password when prompted. Once you have successfully logged in, you can connect the robot's computer to a desired wireless network.

You can connect your robot to a desired wireless network using Netplan.

Simply create a file called 60-wireless.yaml inside of the /etc/netplan folder on your robot's computer. Copy and paste the contents below into the file, and make sure to modify the wireless interface, SSID, and password fields.

```
network:
  wifis:
    # Replace WIRELESS_INTERFACE with the name of the wireless network device, e.g. wlan0 or wlp3s0
    # Fill in the SSID_GOES_HERE and PASSWORD_GOES_HERE fields as appropriate. The password may be included
    # as plain-text
    # or as a password hash. To generate the hashed password, run
    # echo -n 'WIFI_PASSWORD' | iconv -t UTF-16LE | openssl md4 -binary | xxd -p
    # If you have multiple wireless cards you may include a block for each device.
    # For more options, see https://netplan.io/reference/
    WIRELESS_INTERFACE:
      optional: true
      access-points:
        SSID_GOES_HERE:
          password: PASSWORD_GOES_HERE
      dhcp4: true
      dhcp4-overrides:
        send-hostname: true
```

Once you have saved the file, you will then need to apply your new Netplan configuration and bring up your wireless connection by running:

```
sudo netplan apply
```

More advanced networking examples, including configurations for accessing a wifi network requiring WPA Enterprise credentials, can be found here:

<https://netplan.io/examples/>

You can verify that your robot is connected to a wireless network by running:

```
ip a
```

This will show all active connections and their IP addresses, including your robot's connection to the desired wireless network, and the IP address assigned to the robot's computer.

You may continue working on the robot via the wired connection, or disconnect the Ethernet cable and connect to it via wifi instead. When connecting to the robot computer by wifi, it should be accessible via its hostname:

```
ssh administrator@cpr-a200-1051
```

If you are having trouble establishing or maintaining a reliable wifi connection, please contact your network administrator. Some building wifi infrastructures require special permissions to be granted before a connection is allowed.

8 COMMUNICATING WITH THE ROBOT

Ensure the Base Station is turned on. Once it is broadcasting, connect to it via wifi.

To communicate directly with the robot PC, you can SSH in. It will be necessary to ssh into the robot for tasks such as installing, modifying or removing software and files on the robot. If this is your first time connecting to the robot, it will be necessary to connect an Ethernet cable between your own computer and an open port on the robot. Manually set your computer to the same subnet as the robot's IP address.

Note that you will not be able to use GUI tools such as rviz over an SSH connection:

```
ssh administrator@192.168.131.1
```

OR

```
ssh administrator@cpr-a200-1051
```

In order to use rviz and other visualization tools, you must declare the robot as master, and set the user computer IP. In a console on the user pc, type:

```
export ROS_MASTER_URI=http://cpr-a200-1051:11311
```

You should then be able to view a list of topics published by the robot with:

```
rostopic list
```

It will be necessary to declare the robot as master in every new terminal window, unless you change the master permanently in your ROS environment variables. If you are unable to connect with the robot via its hostname, your computer or network equipment may not be routing hostnames properly. In Ubuntu on your local computer, open your /etc/hosts file:

```
sudo nano /etc/hosts
```

Add the following line immediately below the line that contains 127.0.1.1, substituting in the robot's current wifi IP address. This address may be obtained by connecting directly to the robot via Ethernet, and using the "ifconfig" command. You may want to talk to your system administrator about giving the robot a permanent wifi address to ensure it always connects with the same IP address. The below example shows the setting if wired directly into the robot lan.

```
192.168.131.1      cpr-a200-1051
```

To ease communications between the robot and your computer, you can also add a similar entry in the robot's computer, pointing at one or more development computers.

9 SETTING UP YOUR WORKSPACE

To make full use of your new robot, your computer should have a matching version of ROS installed. For best results we recommend using a computer running Ubuntu natively, or in a VM.

Go to <http://wiki.ros.org/noetic/Installation/Ubuntu> for instructions on how to install ROS Noetic on your Linux computer.

Your ROS computer should also have the common software packages associated with your robot model. Once ROS is installed, use apt to install your robot-specific packages:

```
sudo apt-get install ros-noetic-husky-desktop
```

There is a workspace installed in your robot for development and custom packages. You will likely want to be able to use these packages on your local machine. To do this, make a new folder on your local computer:

```
cd ~
mkdir catkin_ws && cd catkin_ws
mkdir src && cd src
```

Then copy all of the packages from the robot to your local machine

```
scp -r administrator@cpr-a200-1051:~/catkin_ws/src/* .
```

Now, make sure you have all of the dependencies needed to build these packages:

```
cd ..
rosdep update
rosdep install --from-paths src --ignore-src -r -y
catkin_make
source ~/catkin_ws/devel/setup.bash
```

10 CONNECTOR PLATE

Your robot is equipped with a custom connector plate. While many of the ports are reserved for payloads on your robot, the remainder are general-purpose ports which may be used to add additional payloads to your system.

The general-purpose power ports supply multiple voltages. See the chart below for pinouts. Please note that the VBat supply is not regulated, and will follow the current battery voltage. When using this supply please ensure that your equipment can handle the full voltage swing of the battery, which can be as much as 23V-30VDC.

Connector	Pin Number	Value	Mating Connector
2-pin EN3P2F16X	Pin 1	12V	EN3C2M16X
	Pin 2	GND	
4-pin EN3P4FX	Pin 1	VBAT	EN3C4MX
	Pin 2	GND	
	Pin 3	GND	
	Pin 4	24V	

Should you need to open the connector plate, do so carefully to ensure that the connected cables are not pulled too tight. When removing the connector plate entirely, disconnect the cables via their in-line connectors, or by disconnecting them from the device to which they are connected.

Power ports and many communications ports contain live power pins when the robot is powered on! Shorting these pins can cause damage to the robot, external equipment, and/or personnel. Exercise caution when connecting and disconnecting devices, and when testing ports.

11 VELODYNE LIDAR

The data produced by the Velodyne LIDAR is best viewed from within rviz, but you can check that it is publishing data using the hz command in ROS. The LIDAR should publish at a rate of about 10Hz or 20Hz, depending on the model:

```
rostopic hz /velodyne_points
```

The Velodyne will also publish a single-beam output to the /scan topic, at the same rate as the pointcloud:

```
rostopic hz /scan
```

Add Velodyne laser data to rviz by navigating to the "Sensing" sub-folder, and adding the relevant Velodyne topic to "LaserScan" and/or "PointCloud2."

12 UM7 IMU

The imu data may be checked by echoing the following topic:

```
rostopic echo /imu_um7/data
```

13 SWIFTNV RTK GPS

The SwiftNav RTK GPS System is a powerful RTK GPS sensor solution. It consists of a Rover-mounted receiver and a fixed-position RTK Base Station receiver. The two receivers are linked via a two-way wifi to serial bridge which permits the Base Station to send RTK corrections to the Rover.

The SwiftNav receiver features a number of status LEDs which can be used to tell at a glance the status of the sensor.

Solid Green	Power On
Flashing Yellow	Waiting for GPS Fix
Solid Yellow	GPS Fix Achieved
Flashing Red	Link Established Between Receivers
Flashing Blue	Floating RTK Solution
Solid Blue	Full RTK Lock

The SwiftNav GPS publishes to /piksi/navsatfix_best_fix. Note that the topic will only publish when a valid GPS fix has been achieved. When a full RTK lock has been achieved, the "Status" fields will change from a "1" to a "2."

```
rostopic echo /piksi/navsatfix_best_fix
```

Additional sensor data from the SwiftNav GPS receiver is available from other topics in the same namespace, such as:

```
rostopic echo /piksi/debug/receiver_state
```

In order to achieve an RTK lock, it is necessary to update the "surveyed position" fields on the base station receiver. To do so, connect to the base station via wifi, and access the base station settings using the Swift Console application. On launch, choose "TCP/IP" and enter the IP address of the base station GPS receiver. Record the current Latitude, Longitude, and Height of the base station receiver (from the Solution tab) and transfer those values to the "surveyed position" fields in the Settings tab. It may be necessary to update these fields every time the Base Station is moved to a new location.

For more information on using the Swift Console, please see this link:

<https://support.swiftnav.com/customer/en/portal/articles/2492795-swift-console>

If the receiver appears to be having trouble achieving a good fix, the following steps may help:

- Make sure the robot has a clear view of the sky
- Move the robot away from tall objects such as buildings and trees
- Make sure no other payloads are partially or fully obscuring the GPS antenna
- Move potential sources of RF interference (such as USB3 cables or powerful radios) away from the antenna and antenna cable

14 VISUALIZING IN RVIZ

You can visualize your robot using rviz. To do so, you must first move a copy of any extra customizations from the robot into a workspace on your local computer (See "Setting Up Your Workspace.") Then, source the workspace:

```
source ~/catkin_ws/devel/setup.bash
```

Declare the robot as ROS Master:

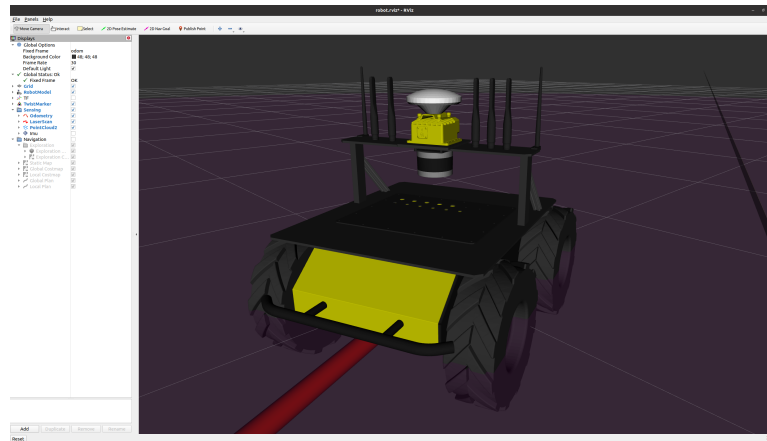
```
export ROS_MASTER_URI=http://cpr-a200-1051:11311
```

To view the robot's model in rviz:

```
roslaunch husky_viz view_model.launch
```

To interact with the robot in rviz:

```
roslaunch husky_viz view_robot.launch
```



You can rotate the model using your cursor, and zoom in or out by scrolling up or down. Strafe by holding down Shift and dragging the model. The robot itself may be driven directly from rviz by changing to interact mode. In this mode, arrows appear around the model. Drag the arrows to make the robot move.

Additional sensor topics may be added to the rviz interface by clicking the “Add” button in the bottom left, selecting “By topic,” then choosing the desired topic from the list.

More information on rviz is available here: wiki.ros.org/rviz

15 ABOUT YOUR INTEGRATOR

This robot was integrated by Zac.

16 LEARNING

If you are new to using ROS, please visit our support page for information on how to get started using your new robot: support.clearpathrobotics.com

Please contact our support team directly at support@clearpathrobotics.com if you have any questions that aren't answered on our support page. For ROS-specific questions, we recommend visiting answers.ros.org, which we also keep an eye on.